

MS4215: Lab 3

Model Building: Systolic Blood Pressure

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Introduction

The dataset `heart_study.csv` contains data from 198 patients in a cardiovascular study. Four missing values in cholesterol are removed before analysis, leaving 194 complete observations. The variables are:

Variable	Description
SYSBP	Systolic blood pressure (mmHg) — outcome
AGE	Age in years
TOTCHOL	Total cholesterol (mg/dL)
GENDER	Sex (1 = Male, 2 = Female)
CURSMOKE	Current smoker (0 = No, 1 = Yes)

Before attempting the questions, run the setup chunk below to load the required packages and data.

```
#install.packages("tidyverse")
#install.packages("car")
library(tidyverse)
library(car)

# In the line of code below, you will need to change the location to where you
# have saved the file
heart <- read_csv("data/heart_study.csv") |> na.omit()

heart$GENDER.f <- factor(heart$GENDER, levels = c('1', '2'),
                        labels = c('Male', 'Female'))
heart$CURSMOKE.f <- factor(heart$CURSMOKE, levels = c('0', '1'),
                          labels = c('Non-smoker', 'Smoker'))
```

Questions

- (i) Explore the data

Use `summary()` to inspect the dataset. Then create two boxplots: systolic BP by gender, and systolic BP by smoking status. Comment on any visible differences between groups.

- (ii) Factor coding

State the reference category for `GENDER.f` and `CURSMOKE.f`. Use `contrasts()` to display the dummy coding for each factor.

- (iii) Fit the full model

Fit a multiple regression model with `SYSBP` as the outcome and `AGE`, `TOTCHOL`, `GENDER.f`, and `CURSMOKE.f` as predictors.

- Which predictors are statistically significant at the 5% level?
- Interpret the coefficient for `AGE` in context.

- (iv) Check multicollinearity

Compute variance inflation factors using `vif()`. Are any a concern?

- (v) Backward elimination

Run backward selection using Akaike information criterion (AIC) (`step()` with `direction = "backward"`), then using Bayesian information criterion (BIC) (add `k = log(n)` where `n = nrow(heart)`).

- Which predictors does each criterion retain?
- Do AIC and BIC agree? If not, which do you prefer and why?

- (vi) Interpret the final model

Based on your selection in (v), state your chosen final model. Write out the fitted equation and interpret each coefficient in the context of blood pressure.